



Embedded Systems Lab

Smart Relay Controller

Weekly Progress Report

Week 3

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Reporting period	Week 3
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1 Executive Summary

Remote relay control over MQTT is working end to end: relays switch on command and the device publishes its state after every change. Two of the four acceptance criteria for the relay-control task are met. The unit test suite now has 27 tests, of which 24 pass; the two failures are in the schedule parser, which cannot yet handle weekday ranges such as Mon-Fri. Fixing the parser and persisting schedules across reboots are the priorities for Week 4.

2 Project Overview

The Smart Relay Controller is an ESP32-based device that switches four relays from a mobile app over MQTT and stores switching schedules on the device. The system has three parts: the ESP32 firmware (PlatformIO), a Mosquitto MQTT broker, and a Flutter mobile app. This report covers the firmware work carried out between 22 and 29 September 2026.

3 Progress Overview

Table 1. Progress by work area

Work area	Status	Progress	Notes
Relay driver	Completed	–	Debounce and brown-out fix merged
MQTT control	Completed	–	Command topic and state publishing
Schedules	In Progress	–	Parser partly implemented; weekday ranges blocked
Relay-control task	In Progress	50% (2/4 criteria)	Acceptance criteria in task record

3.1 Status of previous commitments

Table 2 compares the plan from the Week 2 report with the current state.

Table 2. Week 2 commitments

Commitment	Status	Evidence
Relay driver	Completed	f3009cd, 9b8605b
MQTT control	Completed	e7398c3, bc7b403
Unit tests	Completed	32bf5f9; test-results/junit.xml

4 Work Completed

Table 3. Tasks completed in this period

Task	Status	Evidence
Relay driver with input debounce	Completed	<code>commit f3009cd</code>
MQTT subscription to <code>relay/+set</code>	Completed	<code>commit e7398c3</code>
Relay state published on every change	Completed	<code>commit bc7b403</code>
Brown-out reset when switching relay 3 fixed	Completed	<code>commit 9b8605b</code>
Unit tests for relay driver and schedule parser	Completed	<code>commit 32bf5f9</code>

The brown-out fix removed a device reset that occurred whenever relay 3 was switched, which had made the MQTT connection drop during testing.

5 Work in Progress

The cron-like schedule parser (`commit c5d9452`) parses single days and times. Parsing of week-day ranges and persisting schedules so that they survive a reboot are the two acceptance criteria still open.

6 Results Achieved

Table 4. Unit test results (`test-results/junit.xml`)

Suite	Tests	Passed	Failed	Skipped
relay	18	18	0	0
schedule	9	6	2	1
Total	27	24	2	1

Figure 1 shows the command-to-state latency measured over five runs. The median (p50) stayed between 44 and 51 ms; the 95th percentile ranged from 98 to 135 ms.

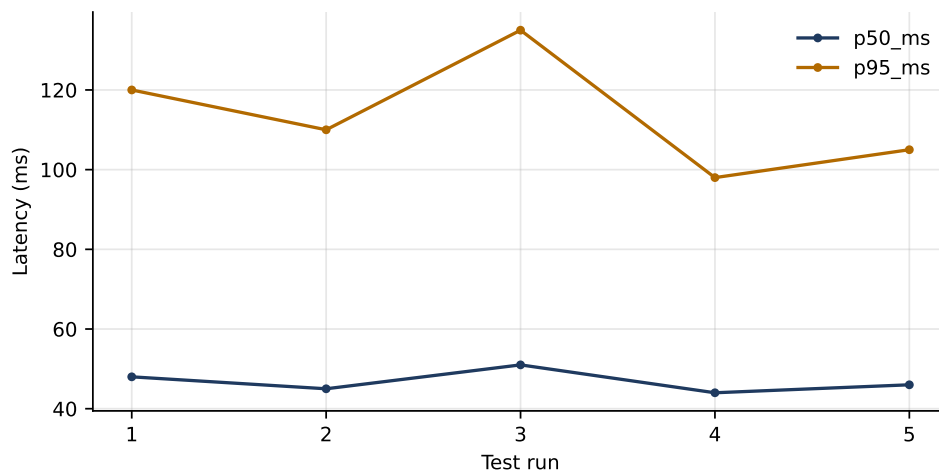


Figure 1. MQTT command latency per test run. Source: `bench/mqtt_latency.csv`.

7 Issues and Challenges

1. **Weekday ranges not parsed.** The tokenizer splits the input on `-`, so `Mon-Fri` is read as two separate tokens (`BLOCKERS.md B-001`). This causes the two failing tests in the `schedule` suite and blocks the related acceptance criterion.
2. **Brown-out reset on relay 3** (resolved this week, see Section 4).

8 Solutions and Corrective Actions

Issue	Status	Action
Weekday ranges	Planned	Proposed: tokenize day fields before splitting ranges, then expand <code>Mon-Fri</code> into a day set
Brown-out reset	Completed	Fixed in <code>9b8605b</code>

9 Next Steps

1. Fix weekday-range parsing and bring the `schedule` suite to zero failures.
2. Store schedules in non-volatile memory so they survive a reboot.
3. Re-run the latency measurement after the schedule module is integrated.

10 Conclusion

All Week 2 commitments were delivered: the relay driver, MQTT control and the first unit tests are in place, and the relay-control task has met two of its four acceptance criteria. The remaining work is concentrated in the schedule module, where one known parser defect is the main open issue.